



User Manual

Release 2.0.0

High-throughput analysis of bacterial ribosome-profiling data

github.com/RickGelhausen/HRIBO

hribo.readthedocs.io

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1 About HRIBO

HRIBO is a reproducible Snakemake workflow for bacterial ribosome-profiling data. It takes gzip-compressed sequencing reads, a reference genome, and a matching annotation and can produce:

- adapter-trimmed reads, filtered alignments, and a combined quality report;
- strand-aware BAM, BigWig, and GFF files for genome-browser inspection;
- feature counts, normalized abundance, and direct translation-efficiency ratios for matched libraries;
- read-length summaries, metagene profiles, and data-driven TIS advice;
- REPARATION and optional DeepRibo ORF predictions;
- PCA, correlation, xTail, RiboRex, and deltaTE results; and
- a combined feature-level overview in spreadsheet, TSV, and GFF formats.

This manual is for people running HRIBO and interpreting its primary results. The complete column-by-column result reference and more detailed biological guidance remain available in the [online documentation](#).

Important. HRIBO 2.0 writes selected results directly into the analysis directory. The result ZIP and the `genome-browser/`, `ORF-predictions/`, and `quality-control/` layouts documented for HRIBO 1.8 are no longer used. Start a 2.0 analysis with fresh configuration and sample-sheet templates.

2 Requirements and installation

2.1 System requirements

The supported platform is Linux on x86-64. Before starting, provide:

- Conda or Micromamba;
- Apptainer installed on the host;
- enough disk space for the data, rule environments, containers, and selected prediction resources;
- a reference nucleotide FASTA and GFF3 or GTF annotation from the same assembly; and
- gzip-compressed FASTQ files.

The first run needs network access while Snakemake creates rule-specific Conda environments and downloads containers, models, or reference data required by the selected stages.

HRIBO's continuous integration tests Apptainer 1.5.3, but the workflow does not enforce that exact version. A working modern installation such as 1.4.5 does not need to be replaced merely to match the test environment. Record the version with the analysis and investigate it if container execution fails.

2.2 Install the 2.0.0 release

Keep the HRIBO checkout separate from every analysis directory:

```
git clone --branch 2.0.0 --single-branch \
  https://github.com/RickGelhausen/HRIBO.git /path/to/HRIBO

micromamba create --name hribo \
  --file /path/to/HRIBO/environment.linux-64.pin.txt
micromamba activate hribo

snakemake --version
apptainer version
```

Use `conda create` instead of `micromamba create` if Conda is preferred. HRIBO runs from its checkout and is not installed with `pip`.

`environment.linux-64.pin.txt` is the reproducible *launcher* environment. It contains Snakemake, Conda, pandas, and the SLURM executor plugin. It is not a monolithic environment containing every scientific tool. Snakemake obtains those dependencies separately from `workflow/envs/` or from the pinned containers when the corresponding rule is needed.

3 Prepare an analysis

3.1 Create a project directory

```
mkdir -p /path/to/my-analysis/{config,data,fastq}

cp /path/to/HRIBO/config/config.yaml \
  /path/to/my-analysis/config/config.yaml
cp /path/to/HRIBO/config/samples.tsv \
  /path/to/my-analysis/config/samples.tsv
```

Copy or link the genome, annotation, and reads into the new project. A small project initially looks like:

```
my-analysis/
|-- config/
|   |-- config.yaml
|   +-- samples.tsv
|-- data/
|   |-- genome.fa
|   +-- annotation.gff3
+-- fastq/
    +-- ribo-control-1.fastq.gz
```

Relative paths in the configuration and sample sheet are resolved from this analysis directory. The FASTA sequence identifier—the first word after `>` in each header—must exactly match annotation column 1, including capitalization and any version suffix.

3.2 Describe the libraries

The sample sheet is tab-separated and has exactly these five headers:

```
method condition replicate fastqFile fastqFile2
RIBO Control 1 fastq/ribo-control-1.fastq.gz ""
RIBO Control 2 fastq/ribo-control-2.fastq.gz ""
RNA Control 1 fastq/rna-control-1.fastq.gz ""
```

```
RNA Control 2 fastq/rna-control-2.fastq.gz ""
```

Leave `fastqFile2` empty for single-end data. For paired-end data, place read 1 in `fastqFile` and read 2 in `fastqFile2`. Single- and paired-end libraries may occur in the same sheet.

Column	Required value
<code>method</code>	One of RIBO, TIS, TTS, RNA, RNATIS, or RNATTS; capitalization matters.
<code>condition</code>	An alphanumeric biological condition such as <code>Control</code> or <code>Treated</code> . Spaces, underscores, and hyphens are not accepted.
<code>replicate</code>	A positive integer. Each method/condition/replicate combination must be unique.
<code>fastqFile</code>	The gzip-compressed single-end FASTQ or read-1 FASTQ.
<code>fastqFile2</code>	The gzip-compressed read-2 FASTQ, or a genuinely empty cell for single-end data.

HRIBO names each library `<method>-<condition>-<replicate>`, for example `RIBO-Control-1`. This name is used throughout the result tree.

3.3 Configure the inputs

Edit the complete copied `config/config.yaml1`. Do not replace it with the short excerpt below, because HRIBO validates the complete section structure. At minimum, check all adapter fields, input paths, alternative starts, and requested stages:

```
biologySettings:
  adapterS3: ""
  adapterS5: ""
  adapterP3R1: ""
  adapterP5R1: ""
  adapterP3R2: ""
  adapterP5R2: ""
  genome: "data/genome.fa"
  annotation: "data/annotation.gff3"
  samples: "config/samples.tsv"
  alternativeStartCodons: ["GTG", "TTG"]

workflowSettings:
  stages:
    - mapping
    - qc
    - tracks
```

The single-end adapter fields are `adapterS3` and `adapterS5`; the four `adapterP*` fields describe paired-end read 1 and read 2. An empty value means no adapter is supplied. It must not be used as a placeholder for an unknown library preparation.

3.4 Choose the results

A stage specifies a deliverable, not an isolated step. Snakemake automatically runs its prerequisites. For example, requesting `predictions` also performs the trimming, mapping, filtering, and counting needed to make those predictions.

Stage	Purpose
trimming	Trim or assemble reads and run raw/processed FastQC.
mapping	Create final unique BAMs after rRNA/tRNA removal.
qc	Build the combined MultiQC report.
tracks	Create raw and normalized strand-aware BigWigs.
genome_tracks	Create possible start, stop, alternative-start, and RBS tracks.
readcounts	Count reads and create annotation/abundance workbooks.
metagene	Summarize read lengths and start/stop-centred profiles.
tis_advisor	Assess mapped read ends, usable lengths, and P-site offsets.
correlation	Compare binned genomic coverage between libraries.
pca	Inspect sample grouping using normalized counts.
predictions	Run REPARATION and, when enabled, DeepRibo.
differential_expression	Run xTail, RiboRex, and deltaTE and build a cross-condition report for valid contrasts.
overview	Combine annotation, abundance, predictions, and enabled differential results.

stages: "preprocessing" selects trimming, mapping, and QC. **stages:** "full" selects every stage, including differential expression. A custom YAML list is usually the clearest choice. A one-off command-line override is also possible:

```
--config stages=mapping,qc,tracks
```

Important. full requires at least two conditions with matched RIBO and RNA libraries, at least two biological replicates on both sides of every contrast, and equal selected RIBO/RNA replicate counts. The overview stage also builds the prediction results it combines and is not a lightweight summary-only action.

When no RIBO library is present, HRIBO removes predictions, differential expression, and overview. Without a RIBO, TIS, or TTS library, it also removes metagene and TIS-advisor work. The resolved stage list is printed at startup.

3.5 Review analysis-specific settings

Prediction. predictionSettings.deepribo accepts on or off. REPARATION is the base predictor. With DeepRibo enabled, the genome may contain only uppercase A, C, G, T, and N. deepriboASiteOffset is measured from a read's 3-prime end to the ribosomal A-site. The template value 12 came from an *E. coli* setup and must be reviewed for the protocol. The TIS advisor evaluates P-site offsets from both read ends and, for RIBO libraries, may also suggest a separate DeepRibo A-site value. This is advice only: compare RIBO libraries before changing the single global setting and rerunning predictions. HRIBO never applies it automatically.

Differential analysis. differentialExpressionSettings.features contains case-sensitive feature types from annotation column 3. If some requested types are absent, HRIBO warns and counts those that exist; if none are present, it stops with an error. A contrast such as Treated-Control is left minus right, so a positive log2 fold change means higher signal in Treated. The filtered workbook sheets use the configured padjCutoff and inclusive log2fcCutoff boundaries. The cross-condition

report uses separate `detectionMinCPM`, `detectionMinCount`, and `detectionMinReplicates` thresholds (defaults: 1 CPM, 10 reads, 2 replicates). A replicate passes detection only when both CPM and raw count meet their thresholds. A feature is detected if enough replicates pass, not detected if none pass and enough usable replicates were available, and uncertain otherwise. A zero-depth sample has undefined CPM and is not a usable replicate. Detection is separate for RNA and RIBO; these settings do not change the statistical tests.

Metagene and TIS settings. Review the allowed read lengths and mapping ends for the library protocol. Metagene filtering and plotting are controlled independently from the TIS advisor. A wide TIS-advisor length range can retain evidence outside the expected metagene range.

4 Run HRIBO

4.1 Validate before computing

Activate the launcher environment, enter the analysis directory, and first construct the complete job graph without running it:

```
micromamba activate hribo
cd /path/to/my-analysis

/path/to/HRIBO/run_hribo.sh \
  --configfile "$PWD/config/config.yaml" \
  --cores 8 \
  --dry-run all
```

The preflight checks configuration keys, sample design, input files, FASTQ structure, reference coordinates, and sequence identifiers. Fix every reported error before starting a long analysis. Listing many planned jobs is normal in a dry-run.

4.2 Run locally

```
/path/to/HRIBO/run_hribo.sh \
  --configfile "$PWD/config/config.yaml" \
  --cores 8 \
  --rerun-incomplete \
  --show-failed-logs all
```

The wrapper selects HRIBO's Snakefile, enables Conda and Apptainer, prints commands, keeps independent jobs running after a failure, waits for filesystem latency, and limits concurrent memory-intensive REPARATION jobs. Additional arguments are passed to Snakemake. Always keep the explicit `all` target at the end.

4.3 Use a larger temporary location

On a workstation or virtual machine, `/tmp` may be much smaller than the analysis filesystem. Before launching HRIBO, point ordinary temporary files and Apptainer's cache/build area to directories with enough space:

```
mkdir -p "$PWD/.tmp/general" \
        "$PWD/.tmp/apptainer-cache" \
        "$PWD/.tmp/apptainer-build"
```

```
export TMPDIR="$PWD/.tmp/general"
export APPTAINER_CACHEDIR="$PWD/.tmp/apptainer-cache"
export APPTAINER_TMPDIR="$PWD/.tmp/apptainer-build"
```

Set these variables in the same shell that launches HRIBO. Confirm that the chosen filesystem has enough capacity and is suitable for temporary data. The `tmpdir=/tmp` text in Snakemake's displayed resources is a temporary-resource assignment; it does not mean every rule writes its large outputs there.

4.4 Resume an interrupted analysis

The normal run command with `-rerun-incomplete` keeps completed outputs and schedules only missing or incomplete work. If Snakemake reports a stale lock, first confirm that no HRIBO process is using the analysis directory, then unlock that exact project once:

```
/path/to/HRIBO/run_hribo.sh \
--configfile "$PWD/config/config.yaml" \
--unlock
```

Never unlock an analysis while another process is active. After unlocking, run the usual command again with `-rerun-incomplete all`.

4.5 Optional SLURM execution

For a SLURM cluster, set the site account and partition in `workflow/profiles/slurm/config.yaml`, activate the launcher environment, and call `slurm_run.sh` from the analysis directory:

```
/path/to/HRIBO/slurm_run.sh \
--configfile "$PWD/config/config.yaml" \
--dry-run all

/path/to/HRIBO/slurm_run.sh \
--configfile "$PWD/config/config.yaml" \
--rerun-incomplete \
--show-failed-logs all
```

Modules, accounts, partitions, and storage policies are site-specific. They are not guessed by HRIBO.

5 Understand the results

HRIBO writes into the analysis directory. Only the selected stages and their prerequisites are present. For a typical analysis, inspect results in this order:

1. Open `qc/multi/multiqc_report.html` and review every library.
2. Check correlation and PCA when multiple libraries are present.
3. Review fragment lengths, metagene profiles, and TIS-advisor evidence.
4. Inspect BAM, BigWig, and GFF files at representative loci.
5. Begin feature-level review with `auxiliary/overview.xlsx`, then return to the method-specific workbook for important candidates.

5.1 Primary result map

Stage	Primary result	Use
trimming	trimmed/<library>.fastq; qc/1raw/ qc/2trimmed/	Mapping-ready reads and FastQC reports before and after trimming.
mapping	maplink/<library>.bam and .bam.bai	Final unique alignments after rRNA/tRNA removal.
qc	qc/multi/multiqc_report.html	Combined read-processing and mapping diagnostics.
tracks	globaltracks/, centeredtracks/, fiveprimetracks/, threeprimetracks/	Raw and normalized forward/reverse BigWig coverage.
genome_tracks	tracks/potential*.gff	Potential starts, alternative starts, stops, and RBS motifs.
readcounts	auxiliary/annotation_total.xlsx, annotation_unique.xlsx, and *_read_counts.xlsx	Feature annotation, RPKM/direct TE, and feature-class summaries.
metagene	metageneprofilng/read_length_ fractions.html and per-library directories	Fragment composition and aggregate start/stop profiles.
tis_advisor	tis_advice/<library>/tis_ recommendation.html	Recommended read end, usable lengths, P-site offsets, confidence, warnings, and separate DeepRibo A-site advice for RIBO libraries.
correlation	figures/heatmap_SpearmanCorr_ readCounts.pdf	Pairwise Spearman correlation of binned coverage.
pca	pca/PCA_3D.html; pca/diffex_QC.html	Interactive sample grouping and count diagnostics.
predictions	auxiliary/predictions_reparation. xlsx, optional predictions_deepribo.xlsx, and tracks/updated_annotation.gff	ORF calls, scores, supporting libraries, and browser annotation.
differential_ expression	diffex_summary/condition_overview. html, diffex_summary/condition_ overview.xlsx, xtail/<contrast>_sorted.xlsx, riborex/<contrast>_sorted.xlsx, deltate/<contrast>_sorted.xlsx	Visual condition detection and contrast changes, browser tracks, and detailed RNA, footprint, and translation-efficiency statistics.
overview	auxiliary/overview.xlsx, .tsv, .gff, and overview_misc.gff	Joined annotation, abundance, predictions, and differential evidence.

5.2 Quality control and final alignments

MultiQC combines diagnostics from raw reads, trimming, mapping, and rRNA/tRNA filtering. Compare read counts, quality, fragment lengths, mapping rates, and depletion across all libraries. A completed report means that the report was generated; it does not mean that the experiment passed biological quality control.

maplink/<library>.bam is a relative symbolic link to the final unique BAM below bam/. Moving the complete analysis tree preserves the link. When exporting BAMs alone, dereference the links and copy each corresponding .bam.bai index.

5.3 Coverage and genome-browser tracks

Each library receives four transcript-oriented views:

global

Coverage over all aligned reference blocks.

centered

Coverage after clipping the aligned read ends.

fiveprime and threeprime

One-position views anchored at the mapped transcript-oriented end.

All four default views are built from the final unique, rRNA/tRNA-filtered alignments. Each is available as **raw**, **mil** (counts per million mapped reads), and **min** (scaled to the smallest mapped library). Forward values are positive and reverse values are negative so a browser can draw the strands on opposite sides of zero; the negative sign is a display convention, not negative abundance.

Use the BAM and raw tracks to judge evidence within one library. Use an appropriate normalized track for cross-library comparison, and load the motif GFFs or **tracks/updated_annotation.gff** alongside them.

5.4 Counts, abundance, and direct TE

annotation_unique.xlsx uses uniquely mapped reads; **annotation_total.xlsx** includes fractional contributions from multi-mapped reads. Both provide feature-level annotation and one RPKM column per library. The smaller **unique_read_counts.xlsx** and **total_read_counts.xlsx** summarize counts by feature class.

When a Ribo-like library has a matching RNA control with the same condition and replicate, HRIBO reports a direct translation-efficiency value:

$$TE_{\text{direct}} = \frac{\text{Ribo-like RPKM}}{\text{matched RNA-like RPKM}}.$$

This is a ratio, not a differential log2 fold change. RPKM values use the mapped total across the complete library, including all contigs.

5.5 Metagene profiles and TIS advice

Start with **metageneprofilng/read_length_fractions.html**, then open each library directory for the configured start- and stop-centred profiles. Check which read lengths contribute, the selected read end, the normalization, the number of included CDSs, and whether the signal fits the experimental protocol.

The TIS-advisor HTML report is the human-readable entry point. Its machine-readable companions are:

- **tis_recommendation.json**, containing the complete result; and
- **read_length_evidence.tsv**, containing evidence for every evaluated length.

Read the recommendation, confidence, warnings, and plots together. A result with no recommended offset can be scientifically valid when the data do not contain a trustworthy initiation peak.

The DeepRibo section converts usable 3-prime-to-P-site estimates to candidate 3-prime-to-A-site offsets by subtracting one codon. It suggests one scalar only when the read lengths agree and represent enough of the RIBO library. Compare that suggestion with **predictionSettings.deepriboASiteOffset** and with other RIBO libraries. Change the global setting and rerun predictions only after review; the advisor does not change it for you.

5.6 ORF predictions

REPARATION calls are stored in `auxiliary/predictions_reparation.xlsx`. DeepRibo adds `auxiliary/predictions_deepribo.xlsx` only when enabled. Review coordinates, prediction score or probability, contributing libraries, sequence, RPKM/TE, and annotation context. Predictor support is evidence to evaluate, not proof of translation.

`tracks/updated_annotation.gff` combines the supplied annotation with accepted prediction calls for browser inspection. Keep the individual workbooks when reporting which predictor supported an ORF.

5.7 Differential results

Open `diffex_summary/condition_overview.html` to scan a searchable, sortable feature-by-condition matrix of RNA and RIBO detection and a separate feature-by-contrast matrix of RNA, RIBO, and translation-efficiency (TE) changes. Click a column heading to sort on its biological state and underlying quantitative value; click it again to reverse the direction. The adjacent `condition_overview.xlsx` contains the same five views in `RNA_detection`, `RIBO_detection`, `RNA_change`, `RIBO_change`, and `TE_change` sheets, with frozen panes, native Excel filters, and the same Feature-plus-matrix layout. Its directional cells remain numeric so Excel sorts their signed log2 fold changes correctly. The adjacent `condition_matrix.tsv`, `contrast_matrix.tsv`, and `browser_tracks.tsv` retain the tabular results. Selected detected features and up/down contrasts are exported as separate `diffex_summary/browser/*.gff3` tracks. Load these alongside the final BAM or BigWig tracks for locus inspection. The report is part of `differential_expression`; it does not require the prediction-dependent `overview` stage.

Detection states are detected, not detected, or uncertain, based on the configured count, CPM, and replicate thresholds. Not detected means below the chosen evidence threshold at this sequencing depth, not proven biological absence. Contrast states are up, down, no directional call, or not tested. The machine value `not\_significant` means that both directional thresholds were not met; it does not establish a high adjusted p-value or zero effect, and is not interchangeable with not detected. The RNA, RIBO, and TE contrast states come from deltaTE; xTail and RiboRex TE estimates are supplemental evidence rather than votes in a combined test. See the [online cross-condition report guide](#) for interpretation and browser-track use.

For `<left>-<right>`, positive log2 fold changes mean higher signal in the left condition. Each xTail, RiboRex, and deltaTE workbook has an `all` sheet and prefiltered up/down sheets. deltaTE separates RNA, RIBO, and TE effects.

Always inspect effect size and adjusted p-value together, using the cutoffs recorded in the configuration. The three tools model translation differently; agreement and disagreement are informative, and similarly named columns are not automatically interchangeable.

5.8 Combined overview

`auxiliary/overview.xlsx` is the most convenient feature-level starting point. Its `all` sheet joins annotated and predicted CDS-like coordinates with sequence, per-library abundance/direct TE, prediction evidence, and enabled differential statistics. Other sheets contain annotated or feature-type subsets.

`auxiliary/overview.tsv` provides the same main table for scripts. `auxiliary/overview.gff` represents the combined CDS set, while `auxiliary/overview_misc.gff` contains non-CDS features. Use the [online result-table reference](#) for current sheet names, dynamic column conventions, coordinates, and missing-value rules.

6 Troubleshooting

6.1 Find the first real failure

Start with the first failing Snakemake rule, not the final downstream summary. The terminal names its rule and normally points to a file below `logs/`. The recommended run command includes `--show-failed-logs`; preserve the complete first traceback when requesting help.

6.2 Preflight stops before any rule

This is intentional. Common causes are a misspelled configuration key or stage, incorrect sample-sheet headers, a path resolved from the wrong directory, malformed or uncompressed FASTQ, mismatched FASTA/GFF sequence identifiers, or a differential contrast that the sample design cannot support. Correct the source input and repeat the dry-run rather than editing generated files.

6.3 A requested annotation feature is absent

Feature types are exact, case-sensitive values from GFF/GTF column 3. If the configuration requests `["CDS", "sRNA"]` but the annotation contains only CDS, HRIBO warns, skips sRNA, and continues with CDS. It stops only when none of the requested feature types occur. Choose feature names that genuinely exist in the supplied annotation.

6.4 Zero or empty biological results

Zero mapped reads are fatal because CPM, RPKM, and normalized coverage cannot be calculated honestly. Work backwards through trimming, reference choice, mapping, and rRNA/tRNA depletion.

By contrast, an all-zero metagene profile, a header-only subset, or a `confidence: none` TIS recommendation can be a valid completed result. Check the rule status, mapped depth, configured lengths and filters, annotation boundaries, and whether the protocol should produce the expected signal.

6.5 REPARATION corrects a P-site estimate

For a sparse or noisy read length, Plastid can estimate a P-site offset that falls outside the read. HRIBO replaces only that impossible estimate with the `default` value in Plastid's table before REPARATION calculates occupancy and ORF predictions. The log names the read length, original value, and replacement, and `p_site_offsets.txt` retains a comment recording the correction. The run may continue successfully because the fallback is valid; inspect `p_site_offset.png` when present and the offset table. Corrections for many well-represented lengths warrant closer review of the signal, annotation, configured footprint range, and library quality.

6.6 Conda or Apptainer fails

Confirm the active launcher environment and record:

```
micromamba activate hribo
snakemake --version
apptainer version
```

The bundled wrappers enable both Conda and Apptainer. The first execution may need network access and considerable cache space. If storage is limited, configure `TMPDIR`, `APPTAINER_CACHEDIR`, and `APPTAINER_TMPDIR` before launching. Replacing a pinned scientific container with unrelated locally installed tools changes the analysis environment and should not be used as an unexplained workaround.

7 Reproducibility, citation, and help

For every analysis, retain:

- the HRIBO version or commit;
- the exact `config.yaml` and `samples.tsv`;
- Snakemake and Apptainer versions;
- the launch command and resolved stage list;
- primary results and relevant logs; and
- any manual changes to offsets, filters, cutoffs, or generated tables.

If HRIBO contributes to published work, cite:

Gelhausen R, Svensson SL, Froschauer K, et al. HRIBO: high-throughput analysis of bacterial ribosome profiling data. *Bioinformatics*. 2021;37(14):2061–2063. doi:10.1093/bioinformatics/btaa959.

For current instructions, visit hribo.readthedocs.io. To report a reproducible problem, use [the GitHub issue tracker](#) and include the first failing rule, its complete log, the software versions, the effective configuration, and a redacted sample sheet. Do not upload confidential sequencing data.

A Pre-run checklist

- The HRIBO checkout and analysis directory are separate.
- The release and launcher environment report the expected versions.
- The FASTA and annotation describe the same assembly and sequence IDs.
- Every FASTQ is gzip-compressed and referenced from exactly one valid sample-sheet row.
- Adapter settings match the single- or paired-end library preparation.
- The requested stages match the available library types and study design.
- Differential feature types occur in annotation column 3 and contrast direction is recorded.
- DeepRibo’s offset and reference alphabet have been reviewed when it is enabled.
- Temporary and cache locations have enough free space.
- The complete `-dry-run all` succeeds before the production command is launched.